

# Steps to use Docker Desktop on Windows Machine with WSL2.

## 1. Enable Virtualization technology (VTx) on your system.

Steps to enable VTx :

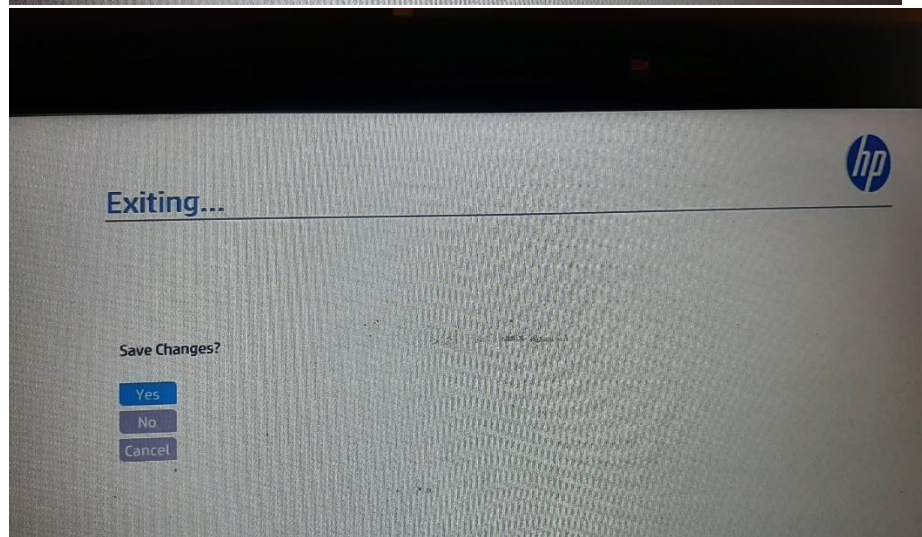
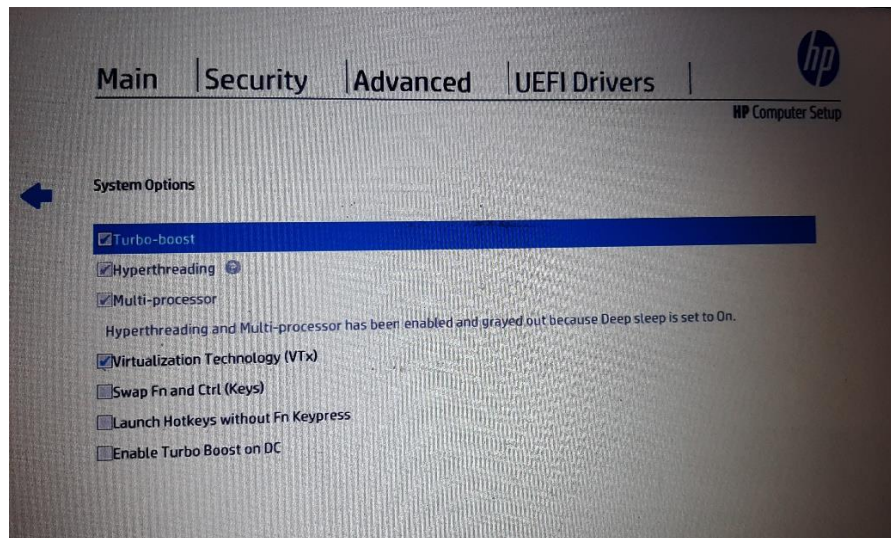
- a) Restart your system and enter "System Setting (BIOS)" by repeatedly pressing ESC/F10 Key depending upon your laptop make and configuration.
- b) Go to Advanced Tab → Select System Options.
- c) Enable Virtualization Technology (VTx)
- d) Save Changes and Exit.



- Display Language
- Scheduled Power-On
- Boot Options
- Secure Boot Configuration
- System Options
- Built-In Device Options
- Port Options ?
- Option ROM Launch Policy
- Power Management Options



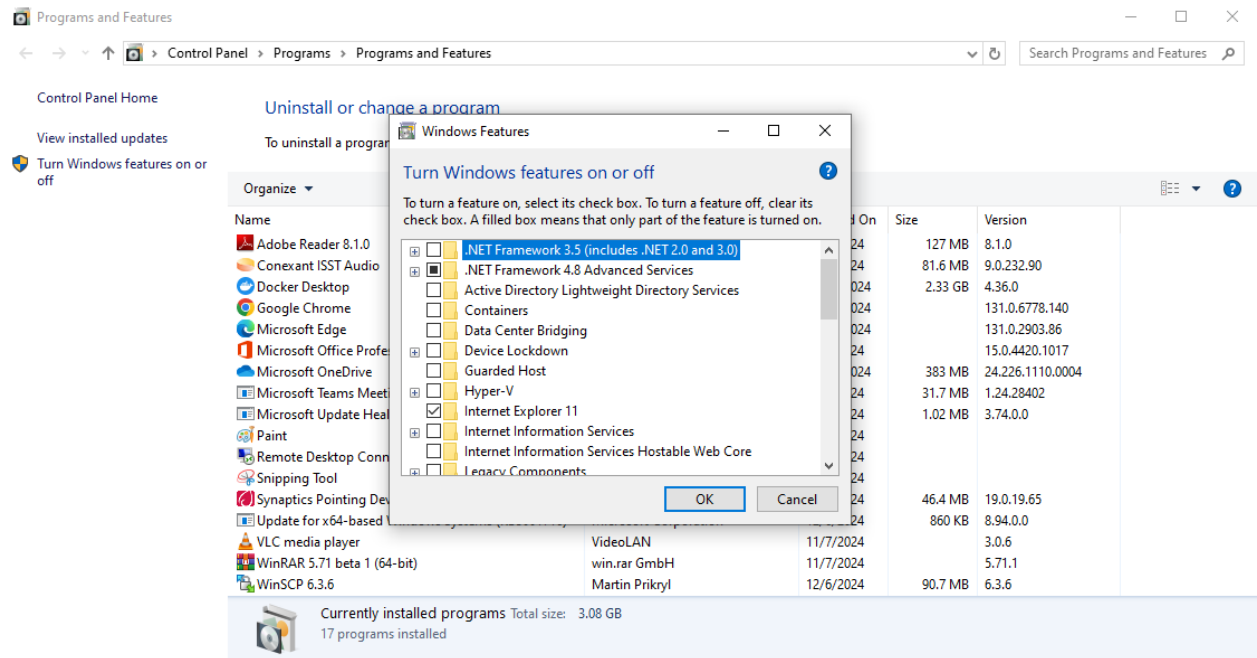
- Display Language
- Scheduled Power-On
- Boot Options
- Secure Boot Configuration
- System Options
- Built-In Device Options
- Port Options ?
- Option ROM Launch Policy
- Power Management Options



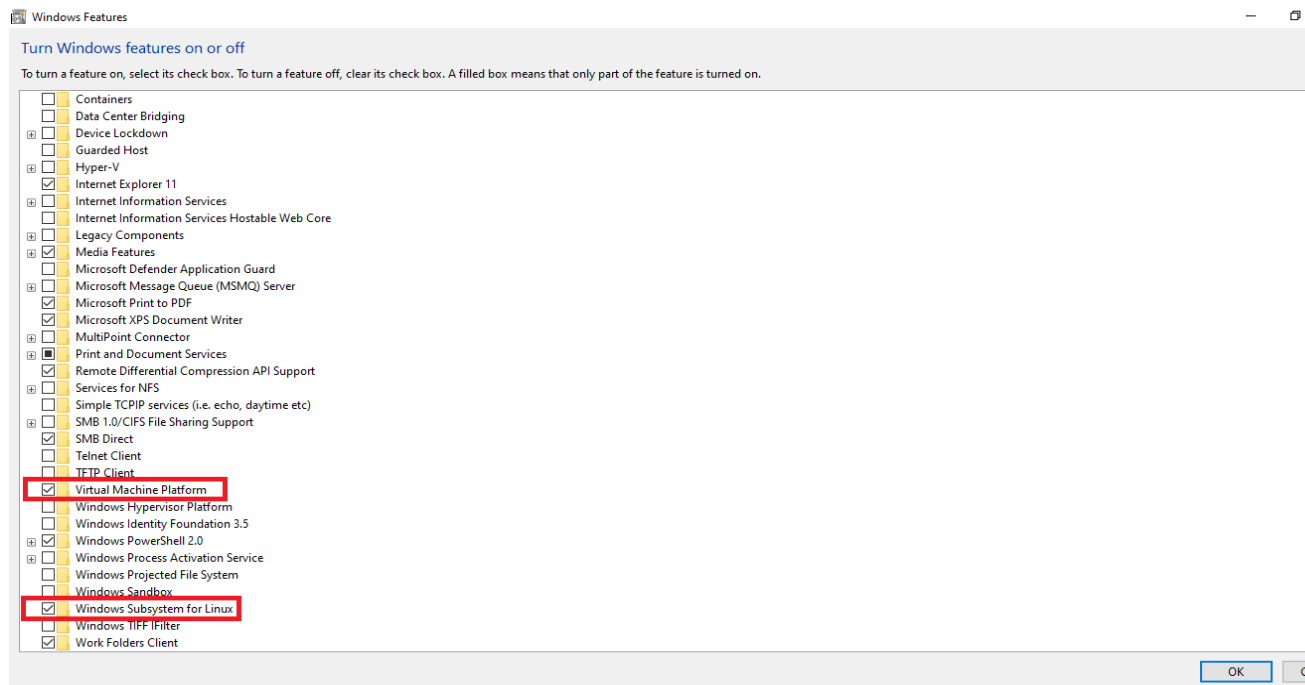
## 2. Enable Virtual Machine Platform and Windows Subsystem for Linux:

Steps:

- Open Settings
- Open Control Panel → Programs → Programs and Features
- Click "Turn Windows features on or off"



d) Check and enable “Virtual Machine Platform” and “Windows Subsystem for Linux”



Click OK to Save and Exit.

### 3. Download and install Docker Desktop with WSL2

Refer docs: <https://docs.docker.com/desktop/setup/install/windows-install/>

#### Download Link of Docker Desktop for Windows – x86\_64:

[https://desktop.docker.com/win/main/amd64/Docker%20Desktop%20Installer.exe?utm\\_source=docker&utm\\_medium=webreferral&utm\\_campaign=docs-driven-download-win-amd64&gl=1\\*1ne0zgc\\*\\_gcl\\_au\\*NTg2MzQzOTAyLjE3MzM5MDIyMzM.\\*\\_ga\\*MTE0NzA4Njk5Mi4xNzMzOTAyMjMz\\*\\_ga\\_XJWPQMjYHQ\\*MTczNDE3MjY0MC4zLjEuMTczNDE3MjY0My41Ny4wLjA](https://desktop.docker.com/win/main/amd64/Docker%20Desktop%20Installer.exe?utm_source=docker&utm_medium=webreferral&utm_campaign=docs-driven-download-win-amd64&gl=1*1ne0zgc*_gcl_au*NTg2MzQzOTAyLjE3MzM5MDIyMzM.*_ga*MTE0NzA4Njk5Mi4xNzMzOTAyMjMz*_ga_XJWPQMjYHQ*MTczNDE3MjY0MC4zLjEuMTczNDE3MjY0My41Ny4wLjA).

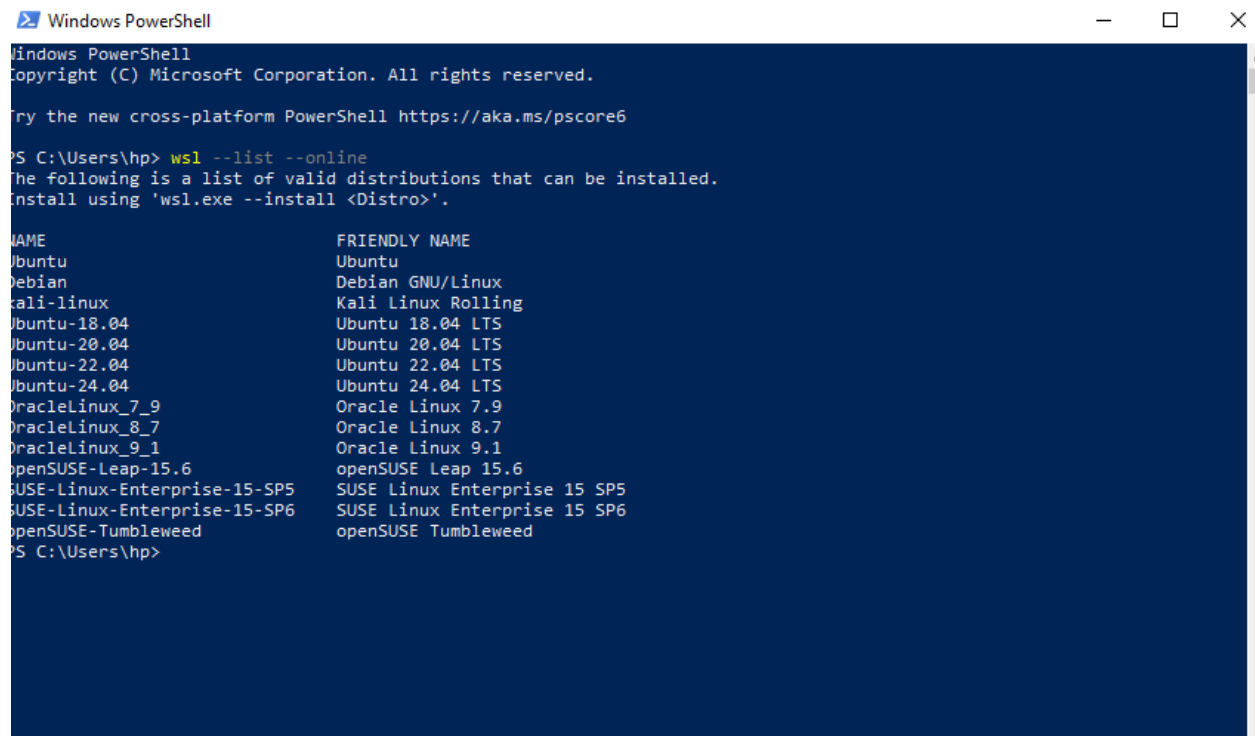
Refer to the system requirement mentioned from the refer docs link.  
Click and install Docker Desktop by selecting WSL2.

#### 4. Install Ubuntu 24.04 LTS on WSL2.

Steps:

a) Open Windows PowerShell/Command prompt and type:

```
wsl --list --online
```



```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\hnp> wsl --list --online
The following is a list of valid distributions that can be installed.
Install using 'wsl.exe --install <Distro>'.

NAME                                FRIENDLY NAME
-----                                -
Ubuntu                              Ubuntu
Debian                              Debian GNU/Linux
kali-linux                           Kali Linux Rolling
Ubuntu-18.04                         Ubuntu 18.04 LTS
Ubuntu-20.04                         Ubuntu 20.04 LTS
Ubuntu-22.04                         Ubuntu 22.04 LTS
Ubuntu-24.04                         Ubuntu 24.04 LTS
OracleLinux_7_9                      Oracle Linux 7.9
OracleLinux_8_7                      Oracle Linux 8.7
OracleLinux_9_1                      Oracle Linux 9.1
openSUSE-Leap-15.6                   openSUSE Leap 15.6
SUSE-Linux-Enterprise-15-SP5         SUSE Linux Enterprise 15 SP5
SUSE-Linux-Enterprise-15-SP6         SUSE Linux Enterprise 15 SP6
openSUSE-Tumbleweed                  openSUSE Tumbleweed

PS C:\Users\hnp>
```

To install a specific version using a NAME from the output type the command as below:

```
wsl --install -d Ubuntu-24.04
```

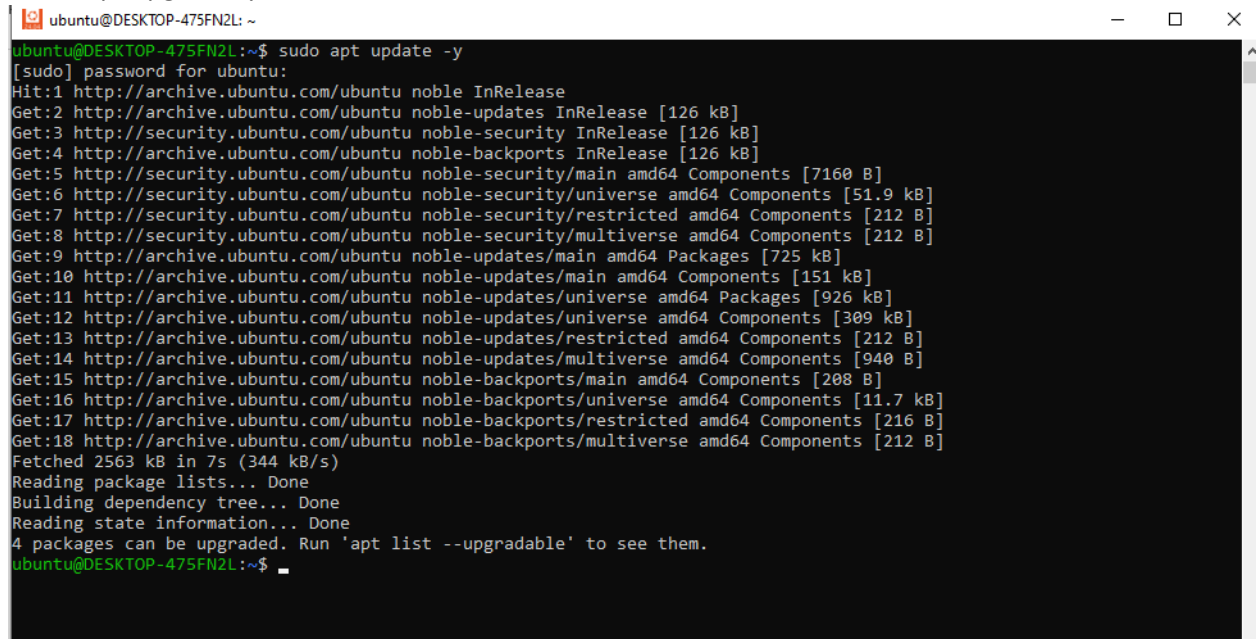
This would install Ubuntu 24.04. Specify the root user name and root password as required.

## 5. Open Ubuntu on WSL2 and update:

Click on Ubuntu 24.04 LTS and update.

```
sudo apt update -y
```

```
sudo apt upgrade -y
```



```
ubuntu@DESKTOP-475FN2L: ~  
ubuntu@DESKTOP-475FN2L:~$ sudo apt update -y  
[sudo] password for ubuntu:  
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease  
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]  
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]  
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]  
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [7160 B]  
Get:6 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [51.9 kB]  
Get:7 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]  
Get:8 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]  
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [725 kB]  
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [151 kB]  
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [926 kB]  
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [309 kB]  
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]  
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]  
Get:15 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [208 B]  
Get:16 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [11.7 kB]  
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]  
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]  
Fetched 2563 kB in 7s (344 kB/s)  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
4 packages can be upgraded. Run 'apt list --upgradable' to see them.  
ubuntu@DESKTOP-475FN2L:~$
```

## 6. Configure Docker and docker compose:

### On Windows:

Open a powershell and type:

```
docker --version
```

```
docker compose version
```



```
Windows PowerShell  
PS C:\Users\hp> docker --version  
Docker version 27.3.1, build ce12230  
PS C:\Users\hp> docker compose version  
Docker Compose version v2.30.3-desktop.1  
PS C:\Users\hp>
```

### On Ubuntu Installed on WSL2:

Docker and docker compose need to be installed:

# Add Docker's official GPG key:

```
sudo apt-get update -y
```

```
sudo apt-get install ca-certificates curl
```

```
sudo install -m 0755 -d /etc/apt/keyrings
```

```
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
```

```
sudo chmod a+r /etc/apt/keyrings/docker.asc
```

# Add the repository to Apt sources:

```
echo \
```



```

"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc]
https://download.docker.com/linux/ubuntu \
$(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
sudo apt-get update -y
sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-
plugin
sudo docker --version
sudo docker compose version

```

```

ubuntu@DESKTOP-475FN2L: ~
ubuntu@DESKTOP-475FN2L:~$ sudo apt-get install ca-certificates curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20240203).
ca-certificates set to manually installed.
curl is already the newest version (8.5.0-2ubuntu10.5).
curl set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 4 not upgraded.
ubuntu@DESKTOP-475FN2L:~$ sudo install -m 0755 -d /etc/apt/keyrings
ubuntu@DESKTOP-475FN2L:~$ sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyr
ings/docker.asc
ubuntu@DESKTOP-475FN2L:~$ sudo chmod a+r /etc/apt/keyrings/docker.asc
ubuntu@DESKTOP-475FN2L:~$ echo \
"deb [> "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://down
load.docker.com/linux/ubuntu \
> $(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
> sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
ubuntu@DESKTOP-475FN2L:~$ sudo apt-get update -y
Get:1 https://download.docker.com/linux/ubuntu noble InRelease [48.8 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [16.6 kB]
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:6 http://security.ubuntu.com/ubuntu noble-security InRelease
Fetched 65.4 kB in 3s (25.6 kB/s)
Reading package lists... Done
ubuntu@DESKTOP-475FN2L:~$ sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plug
in docker-compose-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  docker-ce-rootless-extras iptables libip4tc2 libip6tc2 libltdl7 libnetfilter-contrack3 libnftnlk0
  libnftables1 libnftnl11 libslirp0 nftables pigz slirp4netns
Suggested packages:
  aufs-tools cgroupfs-mount | cgroup-lite firewallld
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-p
  lugin iptables libip4tc2 libip6tc2 libltdl7
  libnetfilter-contrack3 libnftnlk0 libnftables1 libnftnl11 libslirp0 nftables pigz slirp4netns
0 upgraded, 18 newly installed, 0 to remove and 4 not upgraded.
Need to get 125 MB of archives.

```

```
After this operation, 449 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 https://download.docker.com/linux/ubuntu noble/stable amd64 containerd.io amd64 1.7.24-1 [29.5 MB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 libip4tc2 amd64 1.8.10-3ubuntu2 [23.3 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 libip6tc2 amd64 1.8.10-3ubuntu2 [23.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 libnftnl1 amd64 1.0.2-2build1 [14.8 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 libnetfilter-conntrack3 amd64 1.0.9-6build1 [45.2 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/main amd64 libnftnl11 amd64 1.2.6-2build1 [66.0 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/main amd64 iptables amd64 1.8.10-3ubuntu2 [381 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/main amd64 libnftables1 amd64 1.0.9-1build1 [358 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/main amd64 nftables amd64 1.0.9-1build1 [69.8 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble/main amd64 libltdl7 amd64 2.4.7-7build1 [40.3 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/main amd64 libslirp0 amd64 4.7.0-1ubuntu3 [63.8 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble/universe amd64 slirp4netns amd64 1.2.1-1build2 [34.9 kB]
Get:14 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-buildx-plugin amd64 0.19.2-1~ubuntu.24.04~noble [30.7 MB]
Get:15 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-cli amd64 5:27.4.0-1~ubuntu.24.04~noble [15.1 MB]
Get:16 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce amd64 5:27.4.0-1~ubuntu.24.04~noble [25.8 MB]
Get:17 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-rootless-extras amd64 5:27.4.0-1~ubuntu.24.04~noble [9588 kB]
update-alternatives: using /usr/sbin/ebtables-nft to provide /usr/sbin/ebtables (ebtables) in auto mode

Setting up docker-ce (5:27.4.0-1~ubuntu.24.04~noble) ...
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...
ubuntu@DESKTOP-475FN2L:~$ sudo docker --version
Docker version 27.4.0, build bde2b89
ubuntu@DESKTOP-475FN2L:~$ sudo docker compose version
Docker Compose version v2.31.0
ubuntu@DESKTOP-475FN2L:~$
```